



*Module
In
Research Technical
Paper Writing*

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COURSE GUIDE

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GRADING SYSTEM

THE PRELIM AND MIDTERM GRADE IS BASED ON THE FOLLOWING:

Class Performance/Progress Report/Lab Activities	-	40%
Written Report	-	30%
Exam/Oral Presentation	-	30%

THE FINAL GRADE OF THE STUDENT IS BASED ON THE GRADE GIVEN BY THE PANEL DURING FINAL ORAL DEFENSE.

Module 1

How to write Chapter 4: Results and Discussion

INTRODUCTION

The Results and Discussion present the data, the most important part of the paper. The whole paper must stand or fall on the basis of the Results. Data results need to be organized properly in order to facilitate correct interpretation and discussion. This module will orient the students on how to organize, present and interpret research data

OBJECTIVES

After completion of this module, the students must be able;

1. To discuss on how to present data gathered from research.
2. To draft data gathered in visual form.
3. To present discussion that summarize the data gathered.

DISCUSSION

Results and Discussion

Definition

The Results and Discussion section contains all the major experimental findings of the study and their analyses, presented in a logical order with text and visuals that complement and supplement the other.

Contents

The Results and Discussion section contains:

- visuals (tables, figures, and/or illustrations) where necessary for clarity and conciseness,
- text that summarizes the data collected and points out highlights of visuals, and
- any appropriate statistical analyses of the data.

Visuals The visuals should:

- highlight an important point and be referred to somewhere in the text.
- be well designed so they are clearly understood without reference to the text.
- not be redundant. Do not present the same results in multiple visual formats, e.g., presenting the same results in both a table and a graph. Choose the best format for presentation. Are shapes and trends more important to the readers, or exact values?

Text The text should:

- summarize the data collected, point out the important features, and connect the results with one another.
- not interpret the results and discuss the conclusions of the results (a trend can be mentioned, but no interpretation or extended discussion occurs in this section).
- not include raw data. This data should be in a table or in an appendix.

Sequence

Prepare your tables or figures before writing the text. The visual representations will help you clarify your own thinking and make it easier for you to write the Results. This will also help reveal whether there are gaps in the data and whether more experimental work should be done.

Tables

Use tables to show large amounts of data (usually numbers) in a small space. If exact values must be listed, a table is normally preferred over a graph.

Rules

- Put table number and title above the table. The word "table" should be all caps (TABLE 2. Chemical and...). Tables should be numbered with Arabic numerals (in the order of appearance in text). Numbering will enable the writer to refer to them quite easily (Table 2 shows that...).
- Place columns to be compared next to each other, if possible.
- Label each column with a column heading. Make the headings clear but concise. Abbreviations may be used, but do not use periods. Capitalize first words in column headings.

- Include units of measure in the headings if appropriate, e.g., Nitrates (mg/L).
- Use horizontal rules, or lines, if needed. Vertical rules are normally not used; columns are defined by spacing.
- Use single spacing for data and headings. In some instances, you may want to use wide space (extra line) to separate groups of data.
- Align numbers in each column on the right. However, if decimal points are used, the numbers should be aligned on the decimal point.
- Use an initial zero before the decimal (0.25).
- Use notes for more extensive explanation of data or headings. Notes are placed below the table and referenced by superscript letters.
- Order the notes to a table in the following sequence:
 - general notes provide information relating to the table as a whole; place the letter reference at the end of the title
 - column (row) notes refer to a particular column (row); place the letter reference at the end of column (row) heading
 - probability level notes indicate the results of tests of significance
- Tables should be placed as near as possible to the discussion in the text.

Figures

Figures are used to convey the overall pattern of the results at a quick glance. The most common figures are graphs, photographs, and diagrams.

Rules

- Number all figures with Arabic numerals in order of discussion in the text. Number figures separately from tables.
- Label both axes on graphs with the variable being measured, the units of measurement, and the scale.
- Place the figure caption and legend below the figure. A caption is a brief but descriptive title. For example, FIGURE 2. Average Nitrate Content of Wells. A legend consists of one or more sentences that describe what is shown in the figure and point out important features. Sometimes a caption and legend are combined. For example, FIGURE 2. Average Nitrate Content of Wells.

The average of all seven readings of the nitrate level for each well is given. The dotted line represents the maximum contaminant level (MCL) which the EPA has established for public drinking water.

- Figures should be placed as near as possible to the discussion in the text.

Examples

Example 1. Automation Mushroom Cultivation Monitoring System; 2018; De La Pena, Dequito, Merjudio

TABLE 2. Manual vs. Automated Mushroom Cultivation

Stages	Manual Process	Automated Process
Preparing the Substrate	✦ Choosing the Mushroom spawn and substrate to use.	✦ Choosing the Mushroom spawn and substrate to use.
Spawning	✦ Creating the mushroom fruiting bag with the substrate storing it for two weeks with the mushroom spawn.	✦ Storing the fruiting bag with the mushroom spawn until it is ready to fruit.
Fruiting	✦ Adjusting the temperature of the substrate by frequently spraying water on the mushroom bag. ✦ Frequently opening an air duct, ✦ Frequently checking the lighting of the system.	✦ Put the mushroom bag in the automatic monitoring machine, the machine will continuously monitor the value of the rea time temperature of the system, opening the humidifying system if necessary to lower the temperature. Regularly opening the exhaust fan to circulate the air inside the machine and providing a lighting device for the system.

Table 2 shows the equivalent automated process for each manual process of the stages of mushroom cultivation. The basic stages are preparing the substrate, spawning, fruiting and harvesting. It is the same for manual and automated process for preparing the substrate and choosing the spawn to use for the cultivation. The automation takes place during the spawning and fruiting stage of the mushroom. With

the automation these stages becomes easier to facilitate.

Example 2. White Cane with Notification System; 2017; Granada, Manago, Mandocdoc

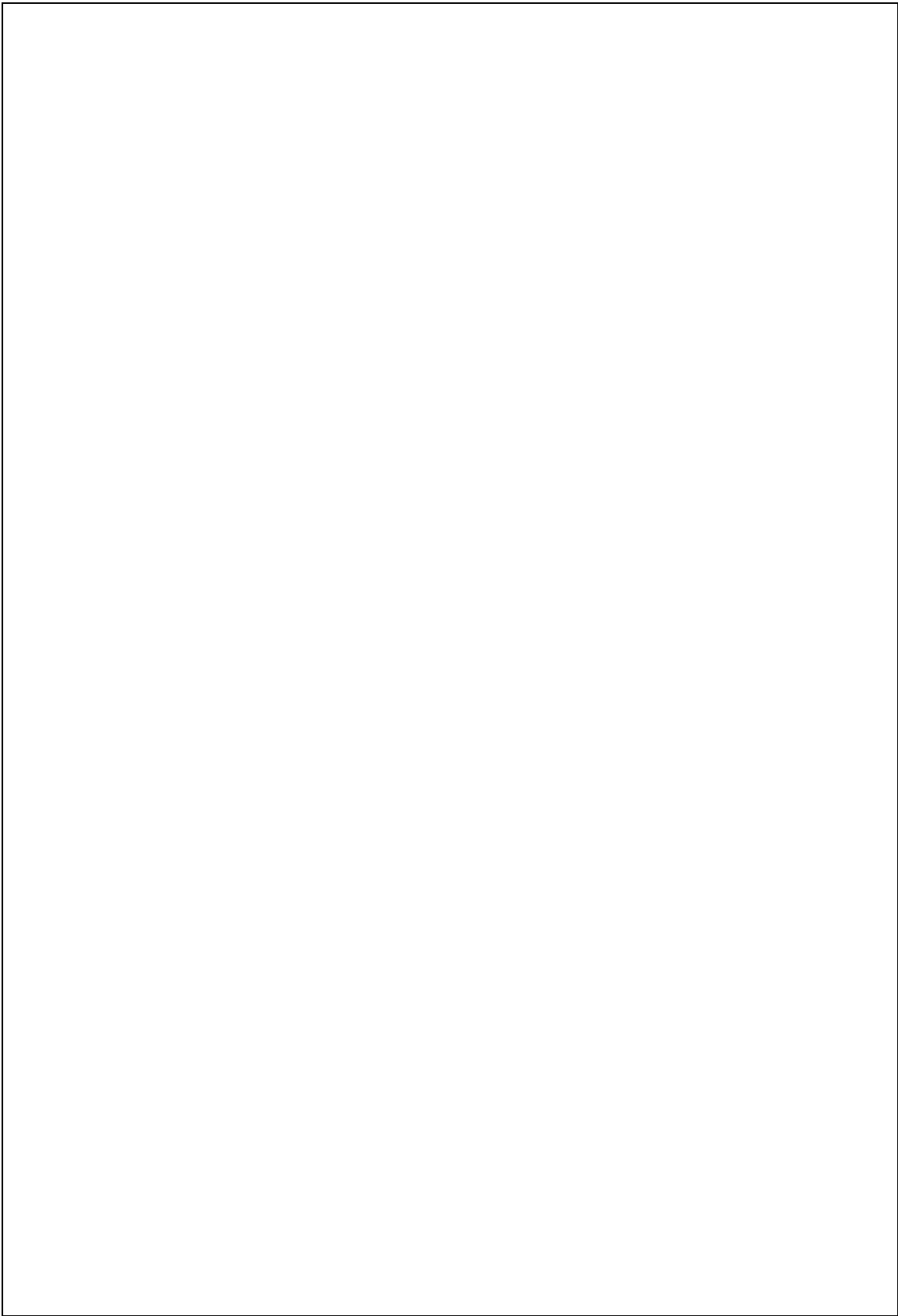


FIGURE 15. Complete Circuit Appearance (B)

The USB ports of the Raspberry Pi are for the broadband stick and the GPS module. The GPS module have been installed into the system through using the USB-TTL.

LABORATORY ACTIVITIES

- 1. Write at least 2 data presentation based on your research project.



REFERENCES

Adelaida B. Almeida, Amelita A. Gaerlan & Norita E. Manly. Research Fundamentals from Concept to Output, 2016

Nelia G. Prieto, Victoria C. Naval & Teresita G. Carey. Practical Research, 2017

Module 2

How to write Chapter 5: Summary, Conclusion and Recommendation

INTRODUCTION

This is the last chapter of the thesis and the most important part because it is here where the findings, and the whole thesis for that matter are summarized; generalization in the form of a conclusion are made; and the recommendations for the solution of the problems discovered in the study are addressed to those concerned.

OBJECTIVES

After completion of this module, the students must be able;

1. To reflect on the guidelines in writing Summary, Conclusion and Recommendation.
2. To create draft of Summary, Conclusion and Recommendation.

DISCUSSION

Summary Of Findings

Guidelines in writing the summary of findings:

- There should be a brief statement about the main purpose of the study the population or respondents, the period of the study, method of research used, the instrument, and the sampling design. There should be no explanation made.
- The findings may be lumped up all together but clarity demands that specific questions under the statement of the problem must be written to be followed by the findings that would answer it.
- The findings should be textual generalizations, that is, a summary of the important data consisting of text and numbers. Every statement of facts should consist of words, numbers, or statistical measures interpreted into a meaningful statement.

- Only the important findings, the highlights of the data, should be included in the summary, especially those upon the conclusion should be based.
- Findings are not explained nor elaborated anymore. They should be stated as concisely as possible.
- No new data should be introduced in the summary of findings.

Conclusions

The Conclusions section sums up the key points of your discussion, the essential features of your design, or the significant outcomes of your investigation. As its function is to round off the story of your project, it should:

- be written to relate directly to the general objective of the project as stated in the Introduction
- indicate the extent to which the objectives have been achieved
- summarize the key findings, outcomes or information in your report
- highlight the significance or usefulness of your work.

The conclusions should relate to the general objective of the work:

Example

General Objective

The aim of this project is to design a mobile phone tower for a suburban location..

Conclusions

The mobile phone tower presented here can be erected safely in a suburban street.

It requires a total area of no more than 2m² l, and has the following safety features:...

Recommendations

- Recommendations are appeals to people or entities concerned to solve or help solve the problems discovered in the inquiry. They should be based on the findings and conclusions of the study.
- Recommendations may be specific or general or both. They may include suggestions for

further studies.

- They should be in non-technical language
- They should be feasible, workable, flexible, doable, adaptable, An action plan is optional.
 - Requires the presentation of suggestions or course of action relevant to the findings of the study which the end-user can adopt.
 - Recommendations should be made in terms of cost, application, and time-elements on the part of the end-user or future researchers.
 - Each suggestion should be explained concisely.

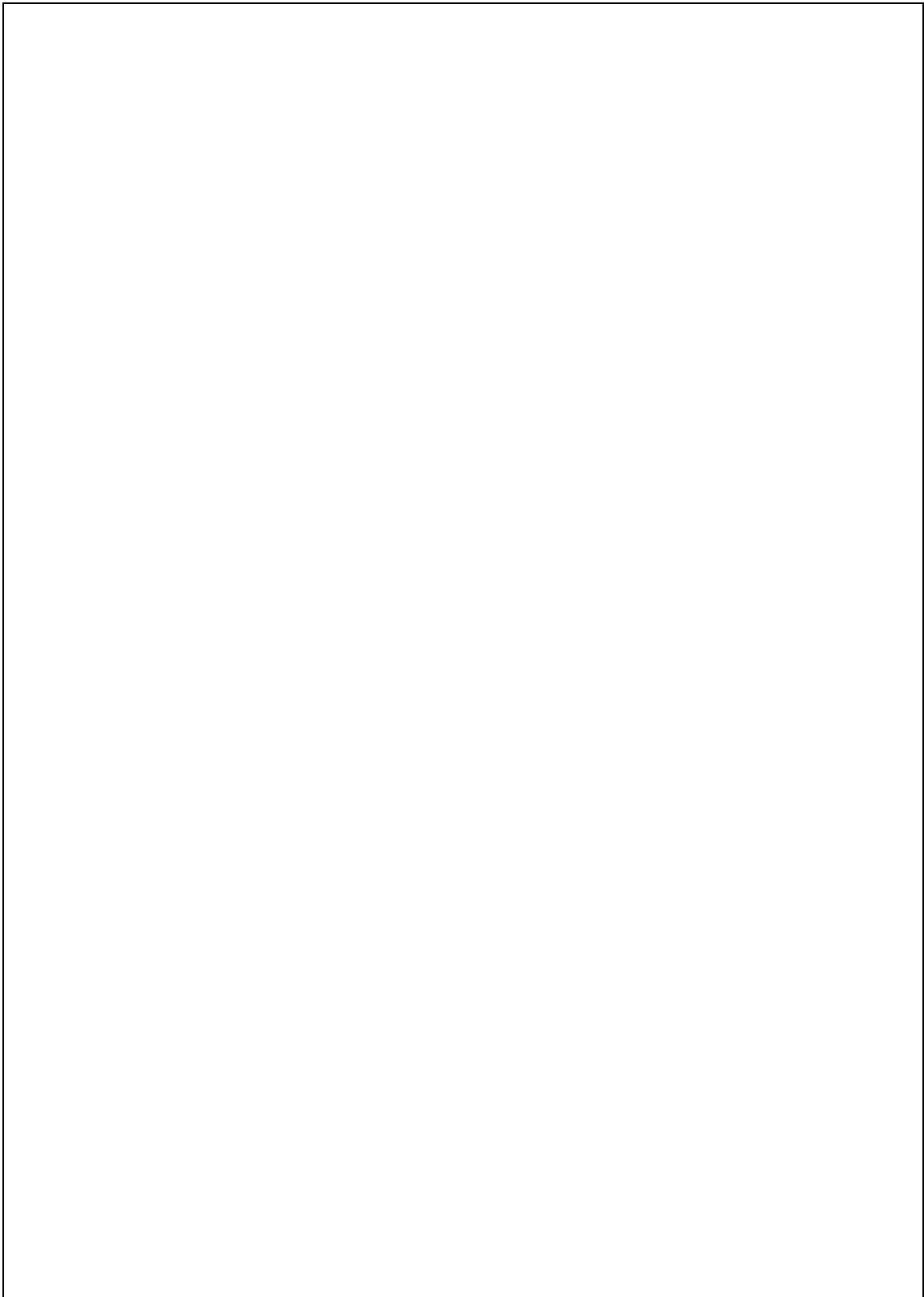
Examples

1. Need for a follow-up study
2. Need for a new study
3. Endorsement of the study to concerned authority

LABORATORY ACTIVITIES

1. Create a draft of Summary, Conclusion and Recommendation based on your Research Project.

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REFERENCES

Adelaida B. Almeida, Amelita A. Gaerlan & Norita E. Manly. Research Fundamentals from Concept to Output, 2016

Nelia G. Prieto, Victoria C. Naval & Teresita G. Carey. Practical Research, 2017

Module 3

Research Project System Overview

INTRODUCTION

A good project system overview provides enough information that the reader has a good sense of the capacity of the system, what it can do, what it can interact with as well as any special requirements. Preparing a comprehensive project system overview requires a detailed knowledge of the system or the ability to get that information from different individuals familiar with the system.

OBJECTIVES

After completion of this module, the students must be able;

1. To reflect on the guidelines in writing Research Project System Overview.
2. To create their Research Project System Overview.

DISCUSSION

How to Write a Project System Overview?

Describe what the system is and what it does. For example, SAP's Project System Overview for Project and Portfolio Management states that it "helps you manage the project through its entire life cycle, from setting up a structure, to drawing up detailed plans, to executing and completing the project." Include what applications apply to specific industries.

Explain how the system works in enough detail for the reader to understand. For example, in the case of the SAP Project System Overview, this includes 14 sub-paragraphs which go into detail from project structuring to producing final reports.

Address any compatibility issues and discuss your system's ability to interact with other

systems or products. For example, in the NASA MEDLI Project System Overview, a slide illustrates for the viewer the "MSL Interface Overview" in order to make the compatibility and interactivity as clear as possible.

State any special requirements for using your product such as hardware or software issues. The special requirements may also be industry-specific. Indicate with a sub-header that these are required in order for your product to work properly. For example, in the NASA MEDLI Project System Overview, there is a special slide dedicated to "Requirements Flow" which is elaborated on in bullets as well as with a flow chart.

Example 1:

PADIBOT: A RICE TRANSPLANTER ROBOT

The design of the prototype is based on hand-cranked manual rice transplanter. The prototype was automated in terms of rice transplanting. It uses a transmitter to control the direction where the prototype can turn, from left to right or vice versa, upon switching on. Arduino Mega 2560 is used as the brain of the prototype in which it is programmed to identify what is toggled in the transmitter then commands the motor to rotate clockwise or counter clockwise. Two DC gear motors, two to drive the wheel drive mechanism for which customized wheels are considered and, are used while one DC gear motor to drive the transplanter mechanism. A 12volt 7Ah batteries are used as a power source that will power all the devices and six 1.5V AA batter for the Arduino Mega 2560.

Example 2:

AUTOMATED MUSHROOM CULTIVATION MONITORING SYSTEM

This system worked to help mushroom cultivation by automating the manual process of mushroom monitoring. The system would use microprocessor for controlling the different materials or devices that was needed in mushroom cultivation. Materials that would be helpful

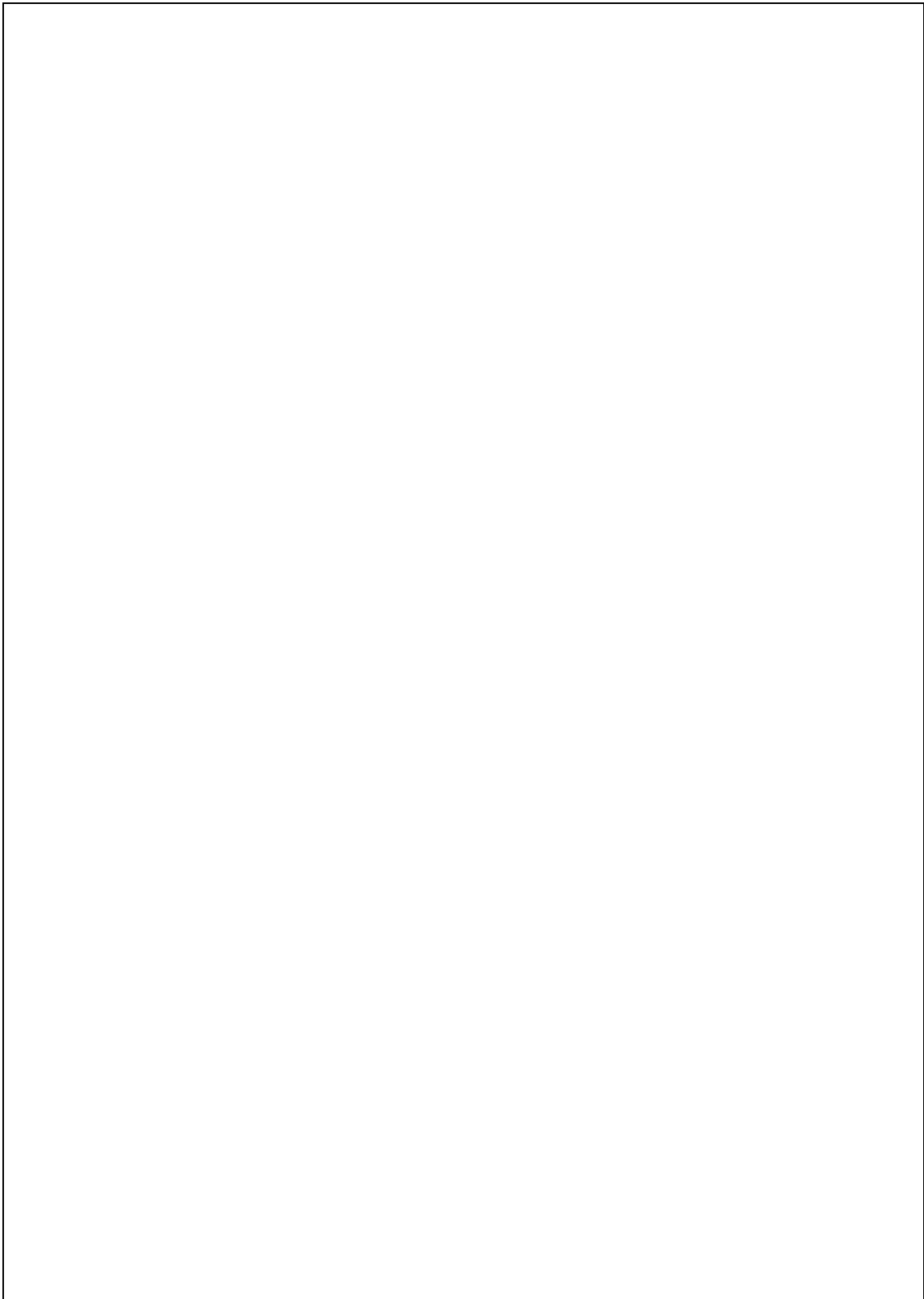
for the mushroom to grow like the needed light and the needed temperature controller devices, were identified.

The system would function as follows: the machine would be equipped with a temperature monitoring device or sensors. This sensor would monitor the real time value of temperature of the system. The system also has an LED monitor that displays the temperature reading of the system. It is programmed in a way that whenever a temperature reaches above a certain point of consideration, the choice of temperature controlling device automatically turns on. The device would create a change in the real time temperature of the system, thus reaching a point that would satisfy the desired temperature for the mushroom to grow. As for the light of the system, a lighting device was equipped inside the machine, providing the necessary lighting for the mushrooms. This lighting device can be turned on and off by the cultivators.

This system helped cultivators by monitoring the temperature automatically hence, decreasing the amount of work load for the cultivators. The system aimed to help the cultivators by providing ease and by helping them to have more free time while the mushrooms are being monitored automatically.

LABORATORY ACTIVITIES

- 1. Write the system overview of your research project.



REFERENCES

Adelaida B. Almeida, Amelita A. Gaerlan & Norita E. Manly. Research Fundamentals from Concept to Output, 2016

Nelia G. Prieto, Victoria C. Naval & Teresita G. Carey. Practical Research, 2017

Module 4

Research Project Technical Description

INTRODUCTION

Technical Description is the second part of chapter 4, Results and Discussion. This includes the plan and design involved in the development of the project.

OBJECTIVES

After completion of this module, the students must be able;

1. To reflect on the guidelines in writing Research Project Technical Description.
2. To present the Technical Description of their research project.

DISCUSSION

Technical Description

Definition

A technical description is text that describes an object or process in terms of its function, organization, parts and details. It is the fundamental "building block" of technical documentation. A technical description can be used alone (e.g., a specification) or as part of a larger document (e.g., a proposal). Often technical documents contain several technical descriptions.

Rules

Typically, a technical description follows a pattern. First, the writer names the object or process and describes its function. Second, the writer gives an overview of the object or process, including its size, parts (with organization of those parts) and other relevant descriptors. Finally, the writer details each significant component, providing explanations, locations and physical descriptions of each component.

A technical description may be short, only a few lines long, or it may be many pages long, giving

details on size, shape, color, material, etc. for multiple components. The writer must determine which elements of a description to emphasize. Visuals are usually included to help the reader understand both the object/process as a whole and its significant elements. Definitions, analogies, and familiar terms or jargon may be used to help the reader.

Content

Technical description presents the process on how did you plan and design your research project.

The following are some presentation of this section whichever is applicable.

- Block diagram
 - Each block of the diagram identifies a component or group of components, their location in the system, the flow of information to and from the components, and the title that identifies the components.
 - Draw and discuss the connection of different components that you used in your project.
 - Example

The planning for the study involved creating the block diagram of the major components of the system and organizing the flow chart that shows the flow of the system program. It also included the circuit design of the components of the system. The block diagram below illustrate the hardware system design for the major components of the prototype.

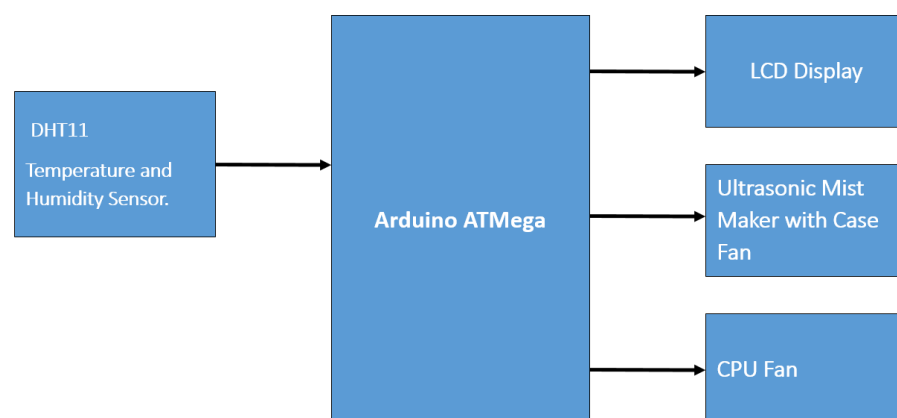
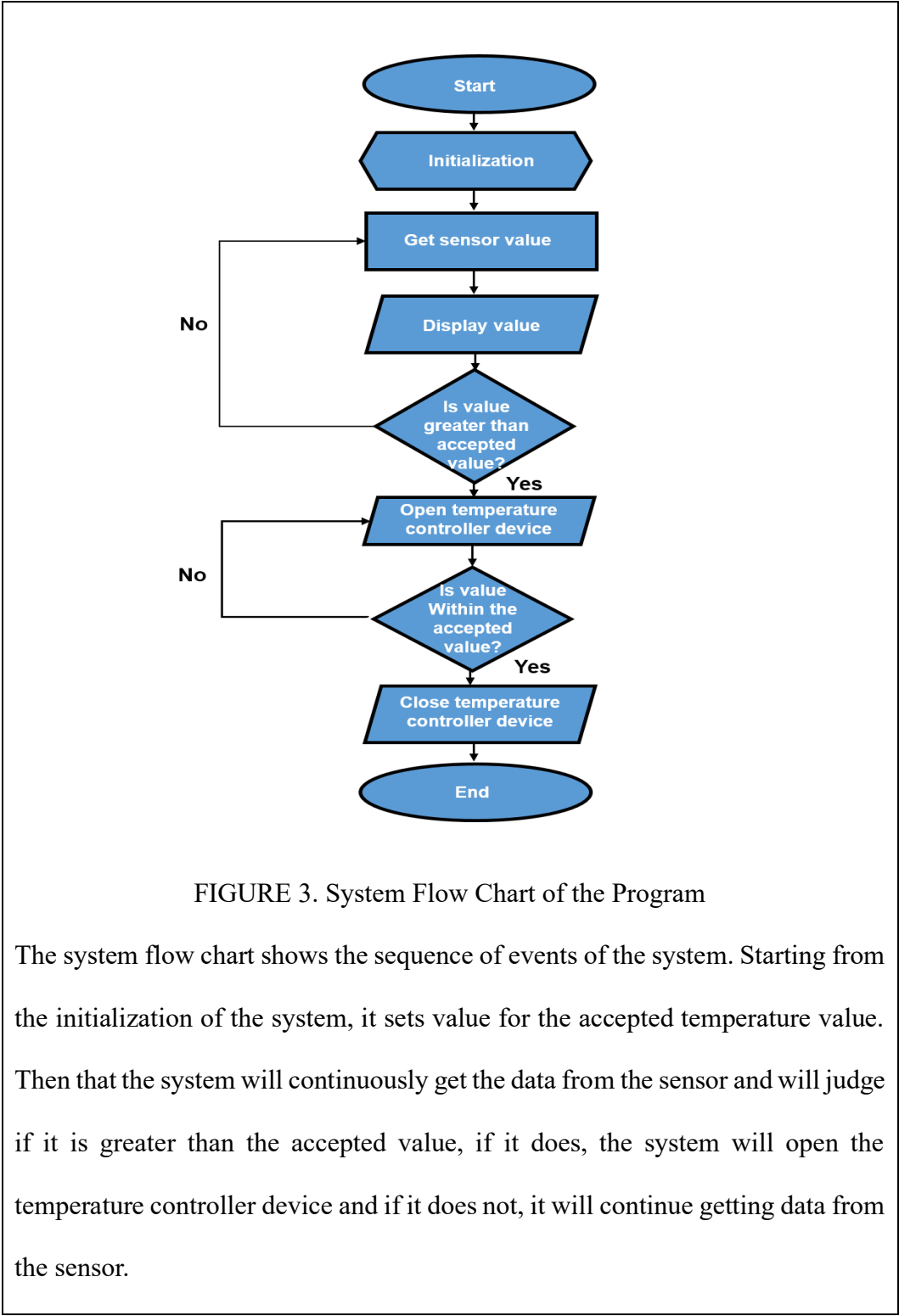


FIGURE 2. Block Diagram

In Figure 2, the sensor get data from the environment and gives these input signal to the microcontroller. When the signal is received, the microcontroller in exchange will give the output signal that will control the ultrasonic mist maker and the CPU fan. The microcontroller can either turn them on or off.

- Flowchart
 - Draw and discuss the data flow of your system.
 - Example



- Circuit Plan
 - Draw and discuss the schematic diagram of the circuit
 - Example

After designing each component circuit, the researcher integrated it all into a one whole system, as shown in the figure below.

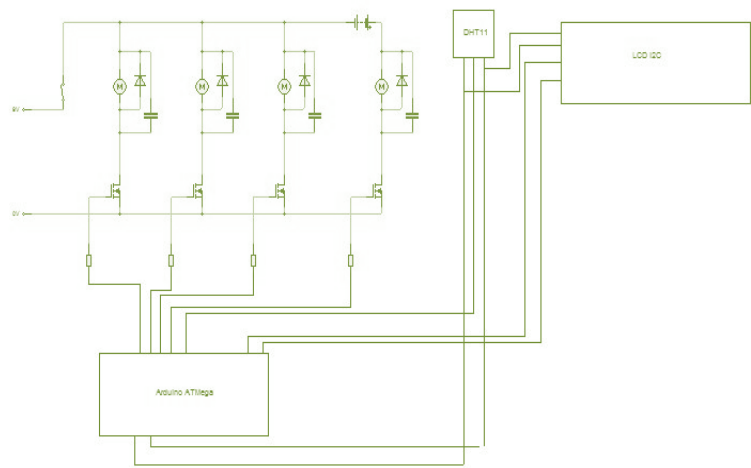


FIGURE 9. Circuit Design of the Whole Circuit

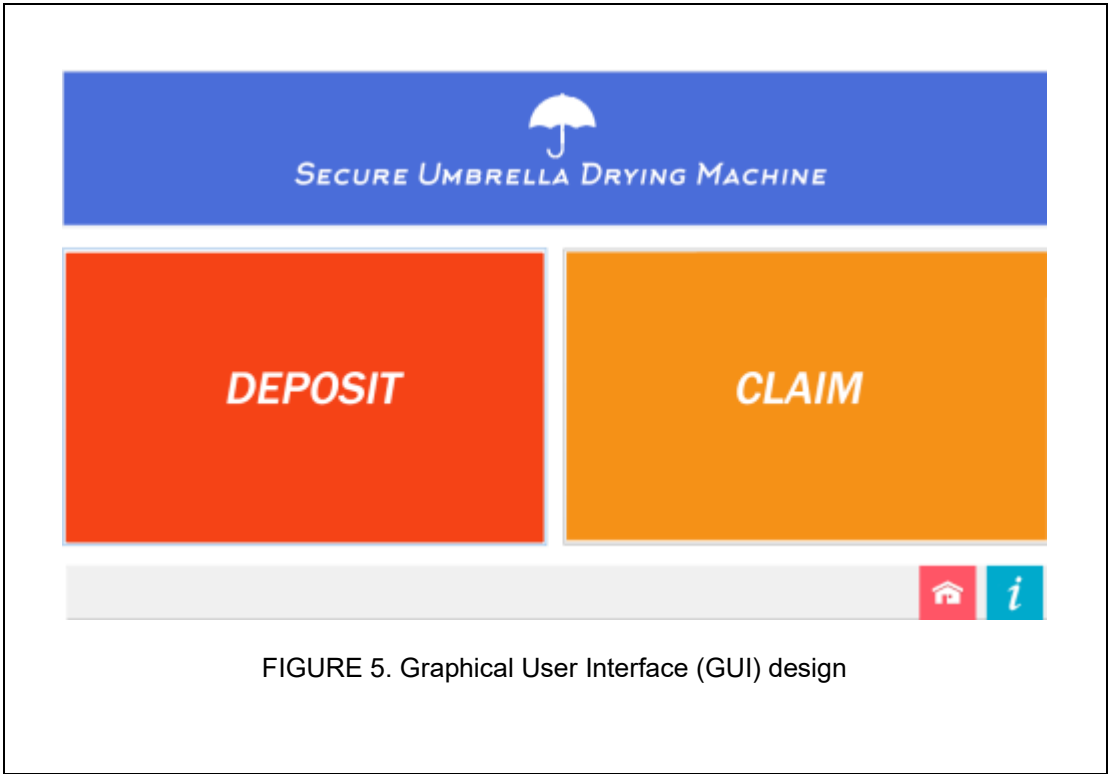
Figure 9 shows the circuit design of the whole system. All the components circuit and their connection to the microprocessor is indicated. The main components of the systems e.g. CPU fan, case fan, ultrasonic mist maker are connected parallel to a 12V source. All are connected to the microprocessor individually through the data pins. The LCD I2C and the DHT11 sensor are both connected to the 5V of the microprocessor and the ground. Their data pins are connected to the analog pin of the microprocessor.

- Materials Used
 - Presents the materials used in the project, its specifications and functions in the system/project.
 - Example

TABLE 4. Main Components for the Automation

Material	Description of Use for the System	Specification
DHT11 	The system needs to monitor the temperature and humidity of the environment. To do this, DHT11 is needed.	• Measurement range 20-90% RH 0-50°C • Accuracy ±5% humidity ±2°C temperature
Ultrasonic Mist Maker 	To control the temperature of the system, the researchers use Ultrasonic Mist Maker, it is a device that uses ultrasonic vibration to create mist from water.	• Mist Propagation: 400 ML / H • Operating Voltage: 12V • Dimensions: 4.6cm Diameter, 4cm Height
CPU Fan 	In mushroom cultivation rate of air exchange is also one thing to consider. The system need to maintain the flow of clean air. In order for clean air to circulate the researchers uses the CPU fan	• Its' dimension is 120 X 120 X 25mm. • Speed of 2,000 RPM. • Rated voltage and current of 12V and 0.32A.

- GUI Design
 - Draw and discuss the prototype case design and its measurement.
 - Example



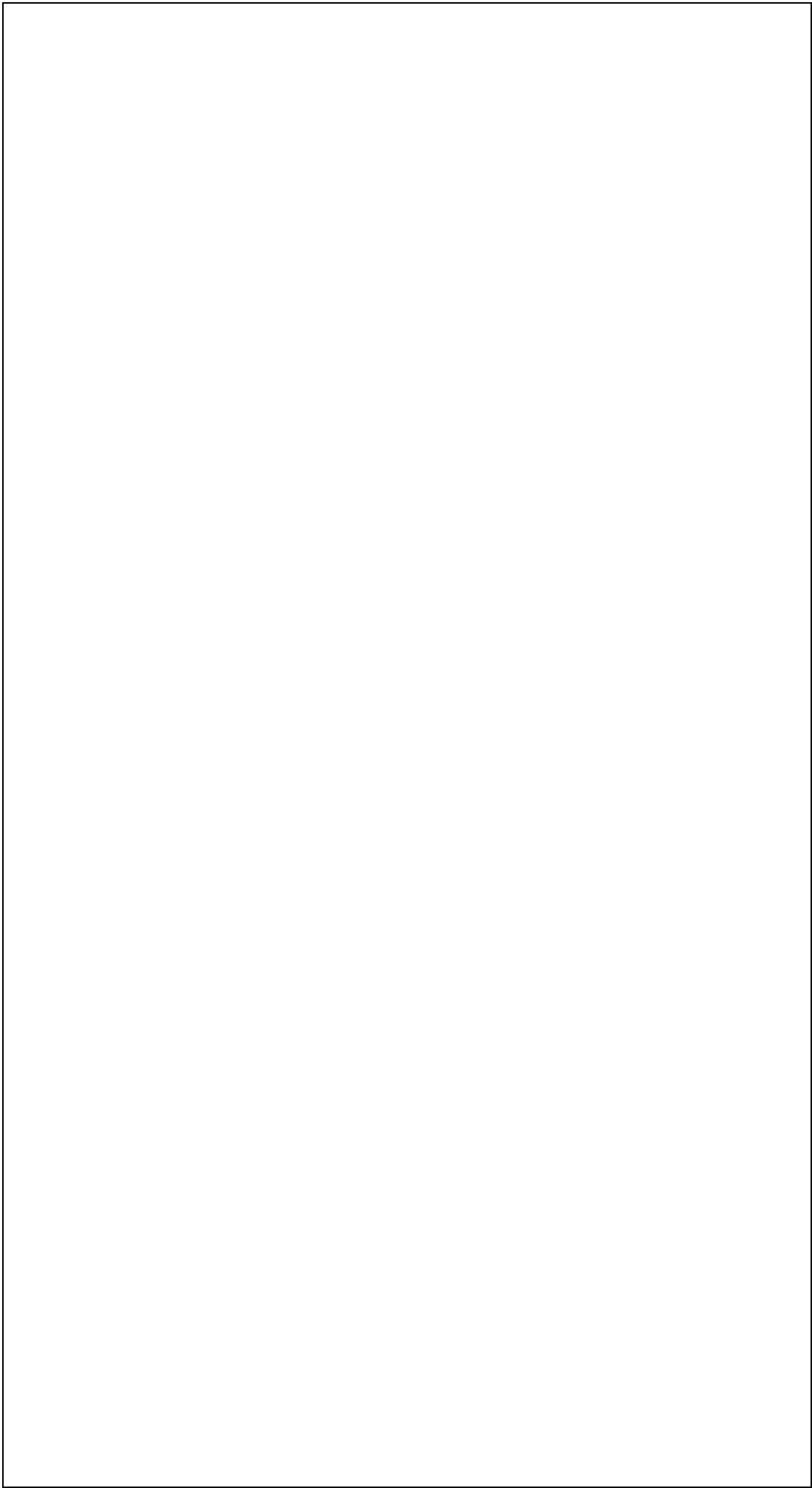
The researchers created a design of the Graphical User Interface. The graphical user Interface will serve as the communication medium for the user to interact with the system. The GUI is composed of buttons for the two transactions namely deposit and claim. Each button corresponds to actions such as getting input from the user using the RFID reader or controlling the servo to release or lock the umbrella slot.

- Case Design
- Etc.

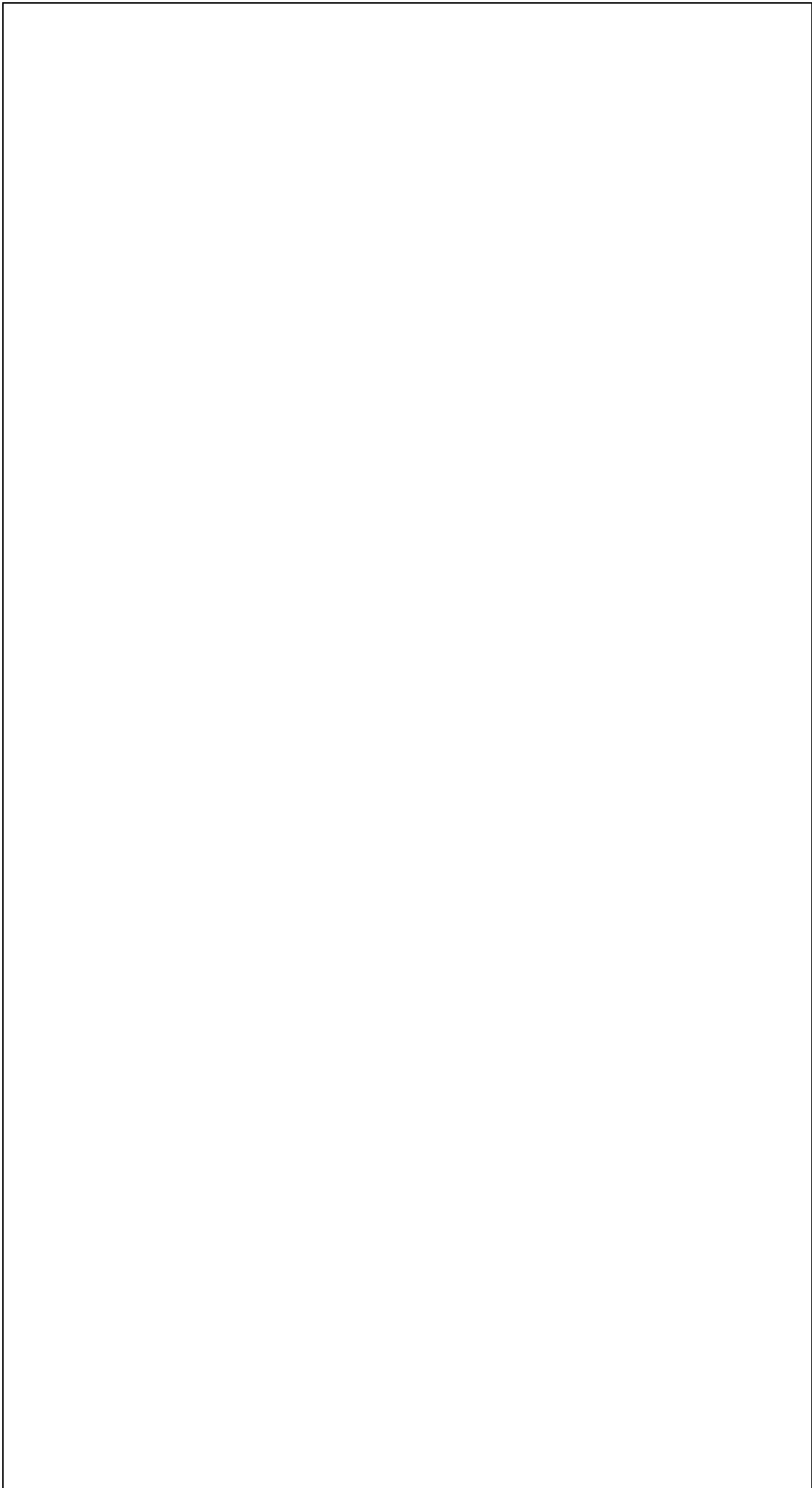
LABORATORY ACTIVITIES

1. Draw and discuss your research block diagram.

2. Draw and discuss your research system flowchart.

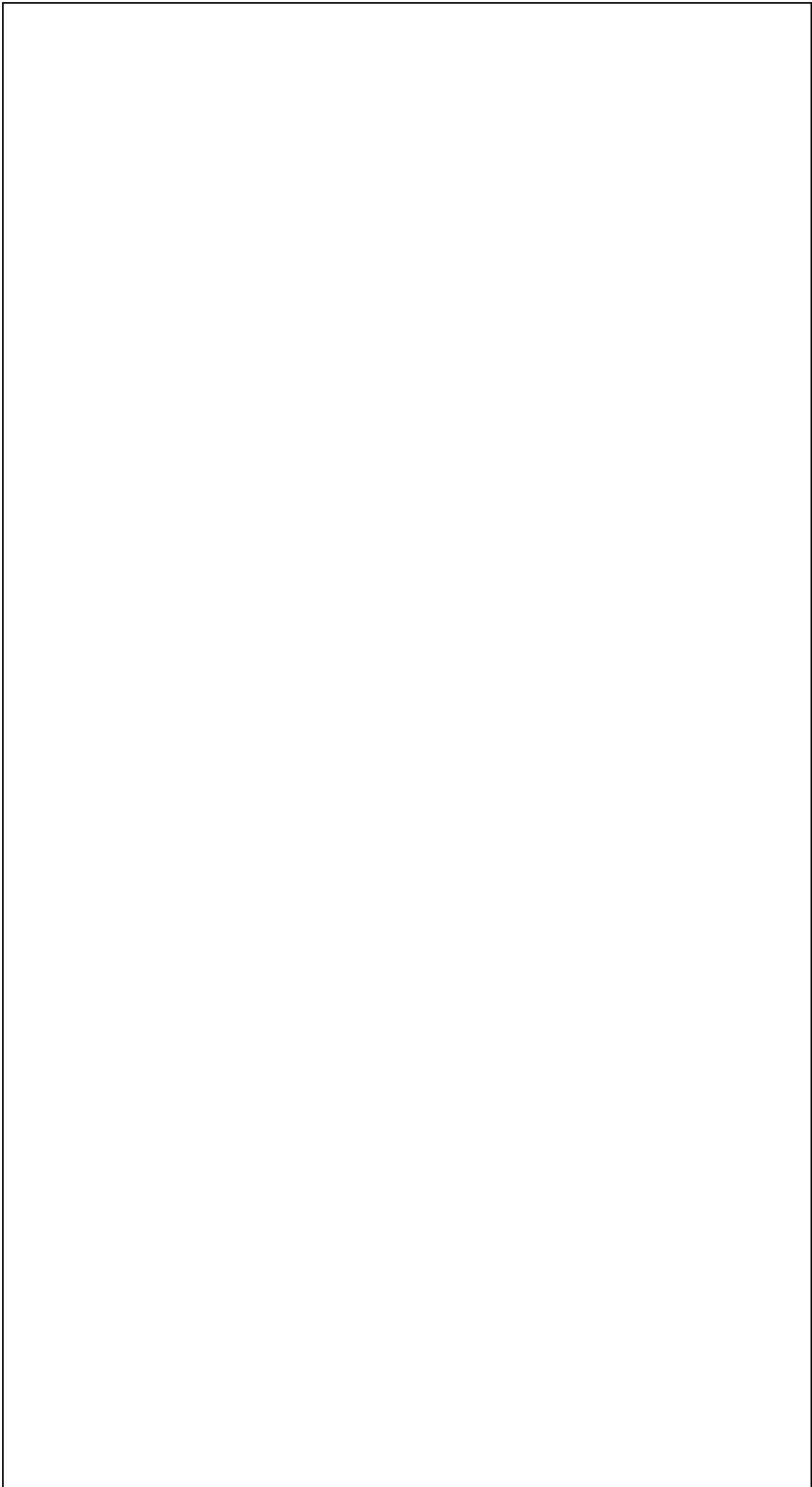
A large, empty rectangular box with a thin black border, intended for drawing a research system flowchart. The box occupies the majority of the page below the instruction.

3. Draw and discuss your research circuit plan.

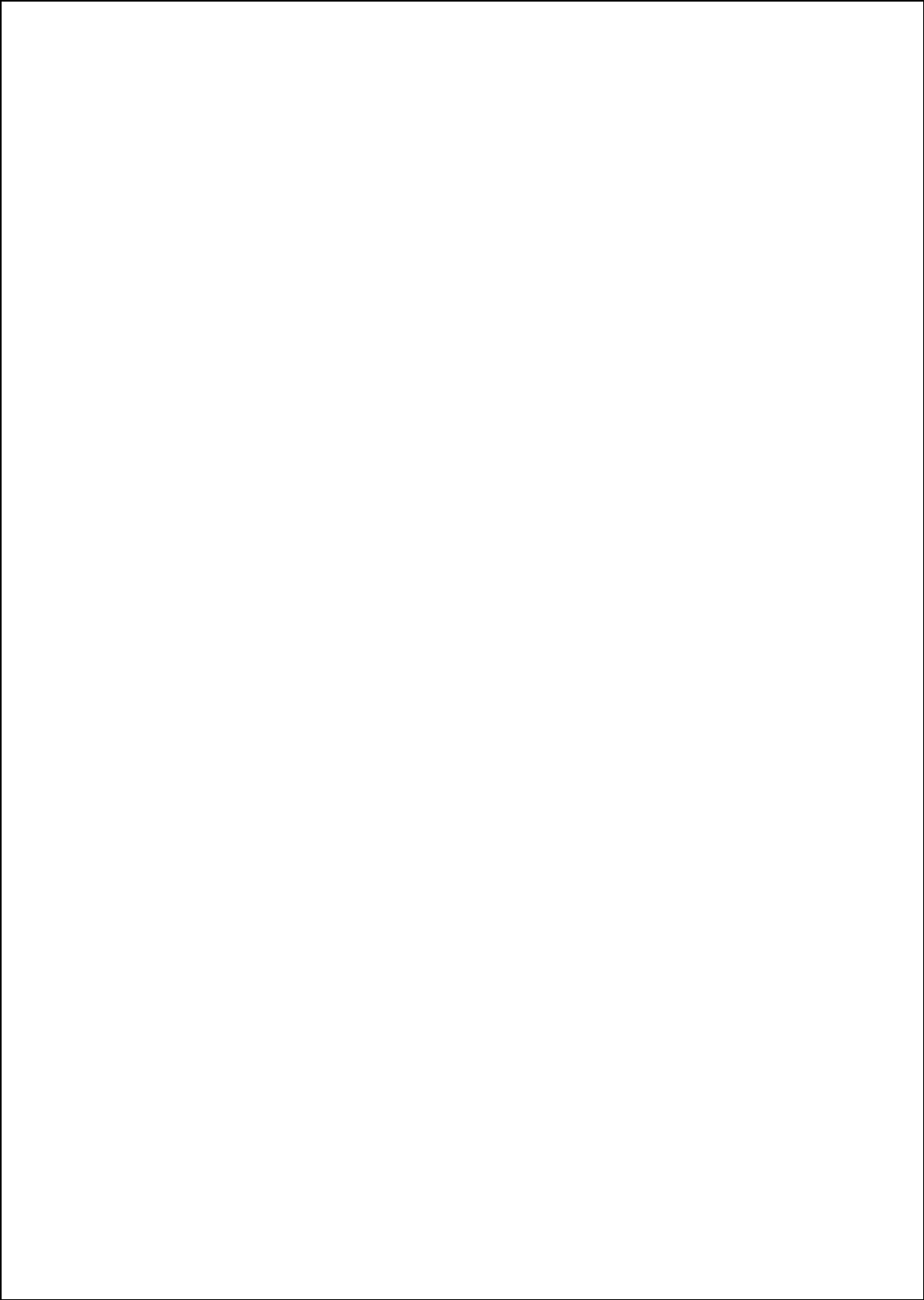
A large, empty rectangular box with a thin black border, intended for a student to draw and discuss their research circuit plan. The box occupies the majority of the page below the instruction.

4. Presents the materials used in your research project.

5. Presents your research GUI design.

A large, empty rectangular box with a thin black border, intended for the student to present their research GUI design. It occupies the majority of the page below the instruction.

6. Draw and discuss your prototype case design.



REFERENCES

Adelaida B. Almeida, Amelita A. Gaerlan & Norita E. Manly. Research Fundamentals from Concept to Output, 2016

Nelia G. Prieto, Victoria C. Naval & Teresita G. Carey. Practical Research, 2017

Module 5

Research Project Structure/Organization

OBJECTIVES

After completion of this module, the students must be able;

- 1. To implement the approved design project.
- 2. To document the construction of approved design project.

DISCUSSION

Structure/Organization section of Chapter 4 covers discussions about the structure or organization of your research project. This includes presentation of program or code, circuit, interfacing of devices, assembling, fabricating and casing of the prototype.

EVALUATION

- 1. Constant monitoring of student research work using the following form:



Republic of the Philippines

SOUTHERN LUZON STATE UNIVERSITY

COLLEGE OF ENGINEERING

Computer Engineering Department



STUDENT RESEARCH MONITORING

Research Title: _____

Name: _____

Assigned Task: _____

DAILY WORK ACTIVITIES

Date	Activities	Remarks:

Monitored by: Engr. Madonna D. Castro Instructor, RES 02		

REFERENCES

Adelaida B. Almeida, Amelita A. Gaerlan & Norita E. Manly. Research Fundamentals from Concept to Output, 2016

Nelia G. Prieto, Victoria C. Naval & Teresita G. Carey. Practical Research, 2017

Module 6

Evaluation

OBJECTIVES

- After completion of this module, the students must be able;
1. To test and evaluate the system/prototype.
 2. To present the evaluation of their Research Project.

DISCUSSION

Evaluation section of Chapter 4, Results and Discussion includes the presentation of your evaluation to your system or prototype. You can evaluate your prototype through survey from the user or through testing. Survey instrument and undergo testing of prototype is used to determine the effectiveness, efficiency, accuracy and reliability of the prototype. The purpose of this is to determine the limitations and capabilities of the prototype.

Example 1. Research Project evaluation through survey

Table 7. Automated Body Temperature Monitoring System’s Design, Effectiveness, and Reliability Survey Results

Criteria	Weighted Mean
<i>Design of the Interface</i>	
1. The design of the interface is pleasing to the eyes.	4.8
2. The graphics/icons of the interface is attention grabbing.	4.7
3. The text is easy to read in terms of font, size and color.	4.9
4. The interface is easy to navigate.	4.6
<i>Effectiveness of Using the System</i>	
1. The device is easy to operate.	4.2
2. The device is able to record the corresponding temperature reading.	4.8
3. The device is safe to use when attached to the patient’s body.	4.8
<i>Reliability of the System</i>	
1. The system can be considered as a device for monitoring body temperature.	4.8
2. The device is a dependable tool in the reading and recording of body temperature measurements.	4.6
3. The device is reliable as a whole.	4.6
Average Weighted Mean	4.68

From the table shown above, the researchers conducted a survey to ten individuals who evaluated the system. The table shows that the average weighted mean results to 4.68. The overall level of the design of the interface, the effectiveness of using the system and reliability gained from the evaluation is Very Satisfactory. The highest average attained was the design in terms of readable texts. However, the lowest average given by the respondents was the operation of the device which attained an evaluation score of 4.2 but it is still within the level of satisfaction.

Example 2. Research Project evaluation through testing

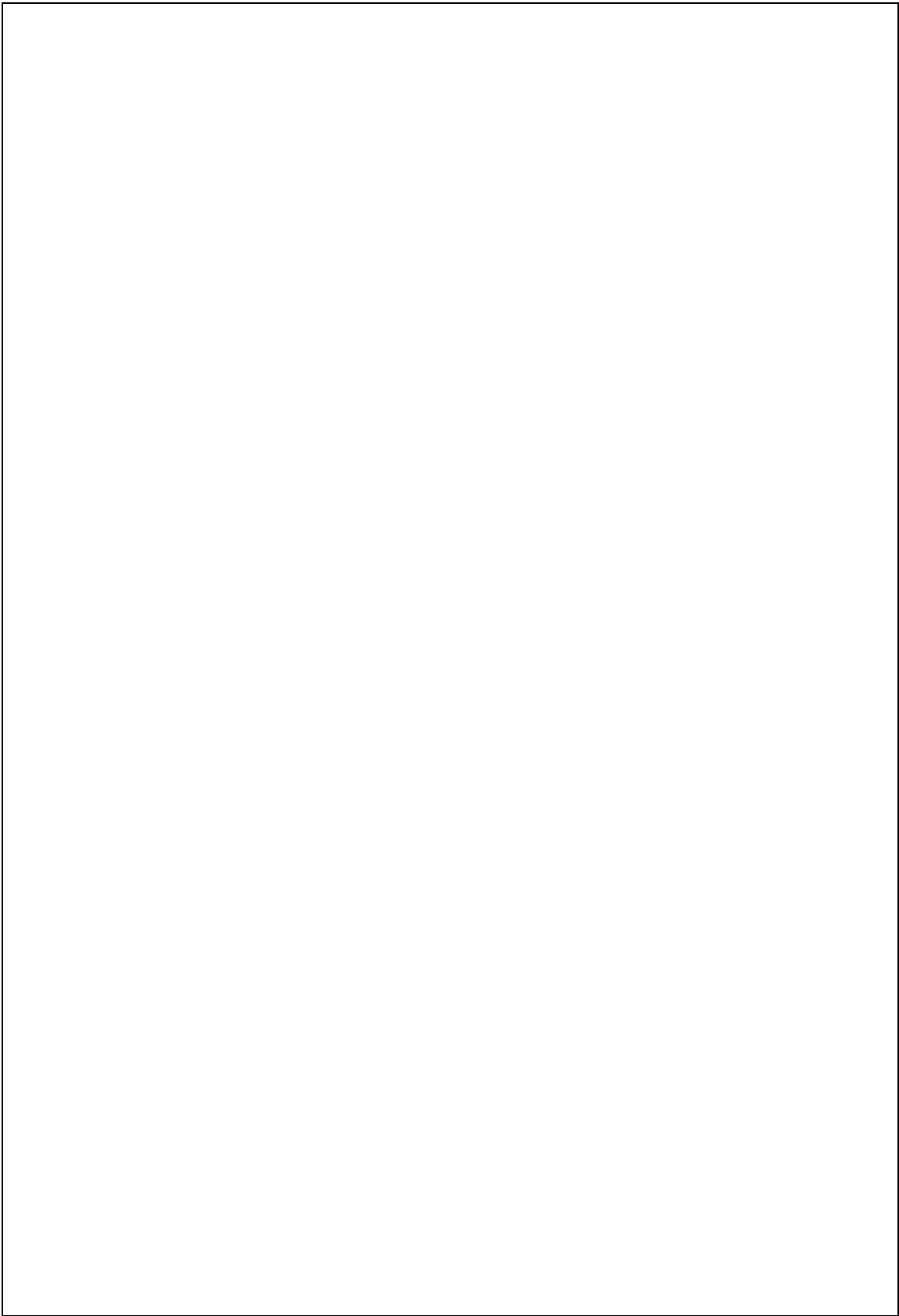
Table 5. Testing the Effectiveness of the Temperature Reading in Different Arm

Trial	Arm Positions		
	Arm Closed (in °C)	Arm Half Inclined (in °C)	Arm Upward (in °C)
1	36.18	36.25	36.06
2	36.06	36.0	35.56
3	36.12	35.93	35.50
4	35.68	35.68	35.25
5	35.87	35.68	35.18
6	35.62	35.37	35.18
7	36.12	36.12	35.93
8	36.06	36.06	35.87
9	35.93	35.87	35.75
10	35.87	35.75	35.68
Average	35.951	35.871	35.596

Table 5 shows the tabularized result of the temperature reading based on different arm positions of the patient. The researchers made ten (10) trials in order to secure the effectiveness of temperature reading on each of the three (3) arm positions which are arm closed, arm half inclined and arm upward with respect to the body. The testing was based on the normal body temperature of an individual in general which is 37°C and the timewas based on a one (1) minute interval for each of the trial. In order to arrive at a precise result, the researchers used the averaging method as basis in the computation. It shows that the nearest average from 37°C was when the arm position is closed which yields to an average of 35.951. Therefore, when using the device, the most applicable arm position in measuring the body temperature is when the arm was closed.

LABORATORY ACTIVITIES

1. Presents at least 3 evaluation of your Research Project.



REFERENCES

Adelaida B. Almeida, Amelita A. Gaerlan & Norita E. Manly. Research Fundamentals from Concept to Output, 2016

Nelia G. Prieto, Victoria C. Naval & Teresita G. Carey. Practical Research, 2017

Module 7

Limitations and Capabilities

OBJECTIVES

- After completion of this module, the students must be able;
1. To enumerate the limitations and capabilities of their Research Project.

DISCUSSION

Limitations and Capabilities section of Chapter 4, Results and Discussion consist of discovered limitations and capabilities of your research project based on the results of testing and evaluation. You can also include, discovered limitations and capabilities during construction of prototype.

Example 1.

Project Limitations and Capabilities

After the evaluation through numerous testing, the researchers found out the capabilities and limitation of system based on the results on the testing.

The system was capable of monitoring the temperature and humidity of the system. The system was also capable of activating the fan and the mist maker when the sensor reads a value of 22°C and was off when the sensor starts to reads 21°C. Also, the system was capable of displaying the reading of the sensor.

The machine was only limited to monitoring the number of mushroom fruiting bag up to 40 pieces. The design of the prototype was designed to house a maximum of 40 mushroom per bag.

LABORATORY ACTIVITIES

- 1. Enumerate the limitations and capabilities of your Research Project.

REFERENCES

Adelaida B. Almeida, Amelita A. Gaerlan & Norita E. Manly. Research Fundamentals from Concept to Output, 2016

Nelia G. Prieto, Victoria C. Naval & Teresita G. Carey. Practical Research, 2017

RESEARCH FORMAT

**TITLE SHOULD BE WRITTEN IN BOLD CAPITAL LETTERS,
12 FONT TIMES NEW ROMAN, DOUBLE SPACED,
INVERTED PYRAMID, CENTERED**

*Margin of the whole
manuscript:
Top: 1”
Bottom: 1”
Left: 1.5”
Right: 1”*

Juan Dela Cruz	<i>Name of authors should be written in bold letters (first name, middle initial, last name) arranged alphabetically.</i>
Juan Dela Cruz II	
Juan Dela Cruz III	

Southern Luzon State University
College of Engineering
May 2020

BIOGRAPHICAL SKETCH

Name: Juan C. de la Cruz

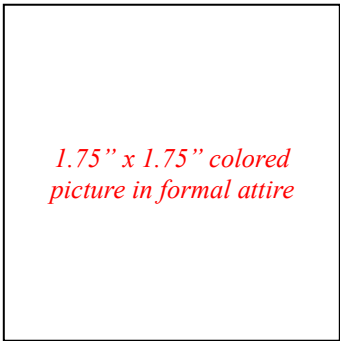
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Educational Background:

School/University	Inclusive Years	Honor/Award
Mababang Paaralan	2003-2009	Valedictorian
Mataas na Paaralan	2009-2013	Valedictorian
University of the Philippines	2013-2018	Summa Cum Laude

Academic Affiliations:

Institute of Electronics Engineers of the Philippines – Quezon Chapter
Member
2016-2018

Future Engineers League
Member
2013-2018

Electronics Engineering Student Society
Member
2013-2017

CERTIFICATE OF ORIGINALITY

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material to which to a substantial extent has been accepted for award of any other degree or diploma of a university or other institute of higher learning, except where due acknowledgement is made in the text.

We also declare that the intellectual content of this thesis is the product of our work, even though we may have received assistance from others on style, presentation and language expression.

Juan Dela Cruz
Researcher

Signature over printed name

Juan Dela Cruz II.
Researcher

Signature over printed name

Juan Dela Cruz III
Researcher

Signature over printed name

Noted by:

Engr. Madonna D. Castro
Adviser

Signature over printed name

May 20, 2020

Date of the oral defense

APPROVAL SHEET

In partial fulfillment of the requirements for the Degree Bachelor of Science in Electronics Engineering, this research paper entitled, **Wifi-connect Printer with Backup Dancer**, has been prepared and submitted by **Juan Dela Cruz, Juan Dela Cruz II and Juan Dela Cruz III**, who is hereby recommended for oral examination.

Title of the study and Names of the authors must be set in bold letters.

Name of adviser, panelists and dean must be in capital letter. Title must be put after the name.

MADONNA D. CASTRO, MEPEE, ICDLSC

Research Adviser

Approved in partial fulfillment of the requirements for the degree, Bachelor of Science in Electronics Engineering, by the oral examination committee.

JERWIN V. OBMERGA, CpE

Member

LEONARD ALLEN R. PAVINO, CpE

Member

CARLA MAY C. CERIBO, CpE

Chairperson

Accepted in partial fulfillment of the requirements for the Degree Bachelor of Science in Electronics Engineering.

May 06, 2020

Date of the oral defense

RENATO R. MAALIW III, PhD

Dean

DEDICATION

To my Mama and Papa double spaced Times New Roman 12 font
Inverted Pyramid not to exceed one page. Author is
allowed a certain degree of creativity
not to exceed one
page

*Initial of the author
without period*

**RDA
RDAJr
RDAIII**

ACKNOWLEDGEMENT

Third person point of view, maximum of 2 pages, 12 font Times New Roman double spaced. There must be no more names/initial of the author after the acknowledgement.

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Abstract

Title: Wifi-connect Printer with Backup Dancer

Authors: Juan Dela Cruz

Juan Dela Cruz II.

Juan Dela Cruz III

Adviser: Engr. Madonna D. Castro

Objectives, Methods, Conclusions, Possible Implications, (Double Spaced, 150-200 words, single paragraph, no indentation) Keywords are also included.

Keywords: (maximum of 5 words, alphabetically arranged, small letter)

Chapter I

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INTRODUCTION

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Tell something about the topic in general. Give facts of information discussing about the topic or problem involved in the study. Show statistics or data to support your topic showing its interesting side and its research ability. USE THIRD PERSON ALWAYS. NEVER USE I, you, us, we. Instead, say the researchers, the world, the people, the nation, they, etc.

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} 3 single spaces before each heading

Background of the study

} 2 single spaces before the start of the paragraph

Tells about how you became interested with this topic for research. Any personal experience, desire, provoking event, felt need, etc. XXXXXXXXXXXXXXXXXXXX XXX XXXXXXXXXXXXXXXXXXXX
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Objectives of the study

The main objective of this technical research is to develop a Wifi-connect Printer with Backup Dancer.

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1. To determine the factors affecting the operation of the prototype.
2. To determine the mechanical and electronics concept of the prototype.
3. To determine the materials needed and suitable components to be used in the research.

Significance of the study

Important contributions of the study to different concerned person or group. XXXXXXXXX
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Scope and Limitation

Include here a) issues, concerns or subject areas of investigation, b) area of coverage, c) number of respondents or subjects involved, d) period of time covered, e) rationale behind delimitating the scope, f) type of data –qualitative or quantitative, g) asks the questions-what, where, and how.

Definition of terms

Start with an introductory paragraph before making your own definition based on how you will use it in your study. Not the meaning found in dictionary. Use your own operational definition kung paano mo ginamit sa study mo. Alphabetically arranged. Sentence form, not hyphenated. Word defined bold.

To make this study understandable to the readers, some terms used by the researchers are hereby defined.

Printer is a machine that is used for printing documents, photograph etc.

Direct current (DC) is an electronic current where the flow of electrons is in one direction only.

Chapter II

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REVIEW OF LITERATURES AND STUDIES

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This chapter presents the related literature and studies of different published books and researches. The use of internet as source of information is considered for better understanding and wider knowledge to this study. Consequently, the obtained information serves as the basis for the conceptual framework.

Related literature

Related studies

Conceptual Framework

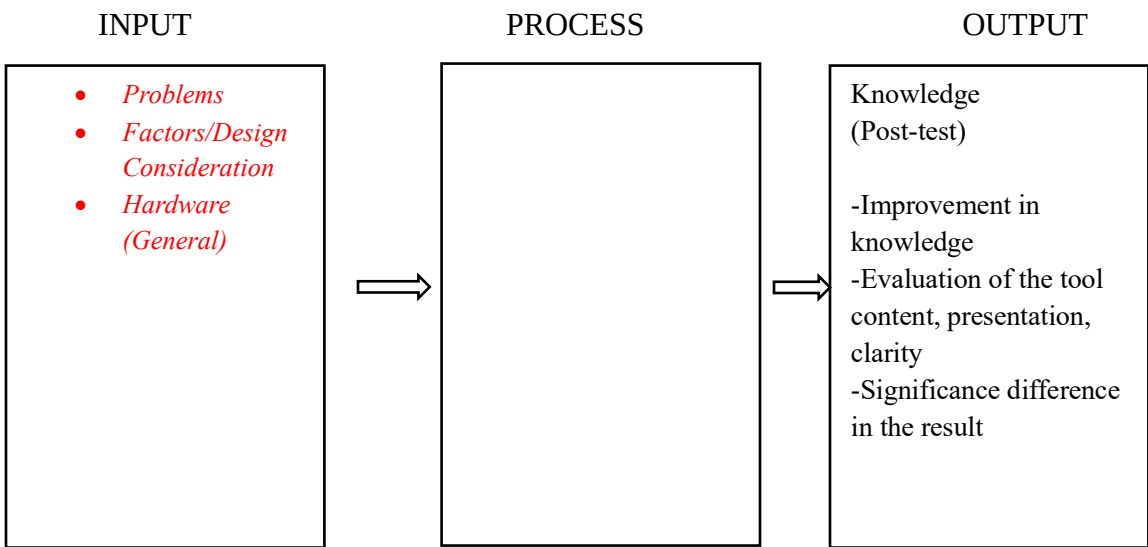


Fig. 1 Conceptual Framework

Hypotheses/Hypothesis (if improvement→Statistical treatment is needed)

Chapter III
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METHODOLOGY
} *3 single space*

This chapter presents the methods and procedures used in conducting the research to utilize and provide the readers an essential view on how the research was made and how the data was collected to develop conclusions. The sources of information are also comprised as reference of the study.

Research Locale

This briefly describes the place where the study was conducted and the rationale behind its choice.

Respondents

The population of interest and the sampling subjects are carefully chosen to suit the purpose of the study so that the validity of the findings will not be jeopardized.

Research Design

Describes the type of research method used. Descriptive, experimental, descriptive/correlational (does not identify cause & effect relationship)

Research Instrument

The nature, purpose, consruction, constant-validation, pretesting, and the rationale for the use of the research instrument/s. it has to be content-validated and pretested for its reliability.

The choices used in preparation of the instrument must be mentioned and how they must be graded or be given equivalent weight for its interpretation. (ex. Use of the 5-pt. Likert scale from Very Satisfactory to Very Unsatisfactory.

Procedures/Data Collection

Contains the step by step procedure used in the administration of the research or experimental instrument applied in the steudy.

Statistical Treatment

Presents and enumerates the appropriate statistical measures and tests used in gathering data. Must mention the justification for the choice of whatever procedure or measure used.

Put the formula when not common. But if the formula used are Weighted Mean, T test, and Chi-square, just mention them; no need to write the whole formula.

Chapter IV**RESULTS AND DISCUSSION***2 single space**3 single space*

This chapter presents the discussion of the acquired data from various tests that serves as the basis of the conclusions and recommendations. It includes the system block diagram, the program flowchart, schematic diagrams and printed circuit board layout, sample computations, and the list of components used. It also deals with the tabulation of the data gathered and their discussions and analysis.

Aside from the tables and graphs, include here photo documentation of your results, findings. Be sure each photo has significant contributions to your findings/results.

Tables/graphs should come in order as they appear in your statement of objectives/problems.

See to it that after the table/graph, the interpretation must follow on the same page as much as possible.

The title of the figure must be written below the diagram while the title of the table must be written above the table. Never cut table. Maximize space in one page or lessen the font to accommodate. Landscape is discouraged.

Always support your findings with literature and findings from similar studies for a meaningful implication.

Explanation about the table is in PRESENT TENSE as if you are talking face-to-face with the reader.

Do not repeat all data written on the table, summarize and give deeper interpretation, not merely repetition.

- System Overview
- Project Technical Description
- Project Structure/Organization
- Project Evaluation
- Project Limitations and Capabilities

Chapter V

SUMMARY, CONCLUSIONS AND RECOMMENDATION

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This chapter presents the summary of the research conducted. It includes the findings gained from the analysis, observation, and interpretation of the results. Recommendations are also formulated for further improvement of the research.

Summary (**Past Tense**)

A brief statement of the following in paragraph form: (not the same as in chapter 1 kasi dapat brief lang. walang subheading ng mga naka enumerate sa baba. In paragraph form nga ano.

- **Main purpose/objective of the study**
- **Population of respondents**
- **Period of the study**
- **Method of research used**
- **Research instrument**
- **Sampling design**
- **There should be no explanation here**
- **Findings** (should answer the specific objectives of the study. It should be in textual generalizations, no detailed data in number, only a summary of important data consisting of text and numbers. Every statement of fact should consist of words, numbers, or statistical measures woven into a meaningful statement. No deduction, nor inferences, nor interpretation should be made, otherwise, it will only be duplicated in the conclusion. Statements must appear in order as in the specific objectives but in paragraph form)

Conclusions (**PRESENT TENSE** dahil generalization na ito ng findings)

(must answer the main objective of the study and the hypothesis/es)

Recommendations (FUTURE TENSE dahil magaganap pa kung sakali. Must be addressed to those mentioned in the significance of the study. Must mention suggestion for concerned person, group, future improvement as found in the findings and for further research)

The researchers hereby recommends the following:

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References Cited

(all documents must be classified as either: books, periodical articles, published & unpublished researchers or retrieved in internet)

Books

Russel, D.E & Norving, P., 2009. *Artificial Intelligence: a modern approach*, 3rd ed., Third Avenue, New York, Mcmillan Publishing Co., Inc..

Journal

Knuth, D.E & Moore, R.W., 1975. An Analysis of Alpha-Beta Pruning. *Artificial Intelligence* 6 (4), 293-326.

Conference Paper

Brin, S. & Page, L., 1998. The Anatomy of a Large-Scale Hypertextual Web Search Engine. In: *Seventh International Conference on World-Wide Web (WWW 1998)*. April 14-18, Brisbane, Australia, Mcmillan Publishing Co. Inc, Third Avenue, New York

Unpublished Work

(accepted for publication but not published)

Creany, N., (in press) *Dummies guide to Professional Ethics*. O'' Really

(circulated informally but not published-Lecture Notes)

Creany, N., 2009. *Lecture Notes on Professional Ethics: [Leaflet] University of Ulster*.

Website

Creany, N., 2008. *Legal issues for IT Professionals* [Online] (Updated 26 September 2008)
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Creany, C., 1959. *Norman at Four Months* [Photograph] (Norman Creaney's private collection).

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Article From an Online Periodical

Author, A. A., & Author, B. B. (Date of Publication). Title of article. Title of Online Periodical, Volume number(issue number if available). Retrieved from <http://www.someaddress.com/full/url/>

Bernstein, M. (2002). 10 tips on writing the living Web. A List Apart: For People Who Make Websites, 149. Retrieved from <http://www.alistapart.com/articles/writeliving>

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Bergmann, P. G. (1993). Relativity. In *The New Encyclopedia Britannica* (Vol. 26, pp. 501-508). Chicago, IL: Encyclopedia Britannica. Work Discussed in a Secondary Source

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Yoshida, Y. (2001). Essays in urban transportation. *Dissertation Abstracts International*, 62,7741A. Dissertation, Published

Lastname, F. N. (Year). Title of dissertation (Doctoral dissertation). Retrieved from Name of database. (Accession or Order Number) Dissertation, Unpublished

Lastname, F. N. (Year). Title of dissertation (Unpublished Doctoral dissertation). Name of Institution, Location. Government Document

Report from a Private Organization

American Psychiatric Association. (2000). Practice guidelines for the treatment of patients with eating disorders (2nd ed.). Washington, DC: Author. Conference Proceedings

Schnase, J. L., & Cunnius, E. L. (Eds.). (1995). Proceedings from CSCL'95: The First International Conference on Computer Support Collaborative Learning. Mahwah, NJ: Erlbaum.

Interviews, Email, and Other Personal Communication

No personal communication is included in your reference list; instead, parenthetically cite the communicator's name, the phrase "personal communication," and the date of the communication in your main text only.

(E. Robbins, personal communication, January 4, 2001).

A. P. Smith also claimed that many of her students had difficulties with APA style (personal communication, November 3, 2002).

Motion Picture

Basic reference list format

Producer, P. P. (Producer), & Director, D. D. (Director). (Date of publication). Title of motion picture [Motion Picture]. Country of origin: Studio or Distributor.

Note: If a movie or video tape is not available in wide distribution, add the following to your citation after the country of origin: (Available from Distributor name, Full address and zip code).

A Motion Picture or Video Tape with International or National Availability

Smith, J. D. (Producer), & Smithee, A. F. (Director). (2001). Really big disaster movie [Motion Picture]. United States: Paramount Pictures. A Motion Picture or Video Tape with Limited Availability

Harris, M. (Producer), & Turley, M. J. (Director). (2002). Writing labs: A history [Motion Picture]. (Available from Purdue University Pictures, 500 Oval Drive, West Lafayette, IN 47907) Television Broadcast or Series Episode

Writer, W. W. (Writer), & Director, D. D. (Director). (Date of broadcast or copyright). Title of broadcast [Television broadcast or Television series]. In P. Producer (Producer). City, state of origin: Studio or distributor.

Single episode of a Television Series

Writer, W. W. (Writer), & Director, D. D. (Director). (Date of publication). Title of broadcast [Television series episode]. In P. Producer (Producer), Series title. City, state of origin: Studio or distributor.

Wendy, S. W. (Writer), & Martian, I. R. (Director). (1986). The rising angel and the falling ape [Television series episode]. In D. Dude (Producer), Creatures and monsters. Los Angeles, CA: Belarus Studios.

Television Broadcast

Important, I. M. (Producer). (1990, November 1). The nightly news hour [Television broadcast]. New York, NY: Central Broadcasting Service. A Television Series

Bellisario, D. L. (Producer). (1992). Exciting action show [Television series]. Hollywood, CA: American Broadcasting Company.

Music Recording

Songwriter, W. W. (Date of copyright). Title of song [Recorded by artist if different from song writer]. On Title of album [Medium of recording]. Location: Label. (Recording date if different from copyright date).

Taupin, B. (1975). Someone saved my life tonight [Recorded by Elton John]. On Captain fantastic and the brown dirt cowboy [CD]. London, England: Big Pig Music Limited.

Article in Journal Paginated by Volume

Journals that are paginated by volume begin with page one in issue one, and continue numbering issue two where issue one ended, etc.

Harlow, H. F. (1983). Fundamentals for preparing psychology journal articles. *Journal of Comparative and Physiological Psychology*, 55, 893-896.

Article in Journal Paginated by Issue

Journals paginated by issue begin with page one every issue; therefore, the issue number gets indicated in parentheses after the volume. The parentheses and issue number are not italicized or underlined.

Scruton, R. (1996). The eclipse of listening. *The New Criterion*, 15(3), 5-13.

Article in a Magazine

Henry, W. A., III. (1990, April 9). Making the grade in today's schools. *Time*, 135, 28-31.

Article in a Newspaper

Unlike other periodicals, p. or pp. precedes page numbers for a newspaper reference in APA style. Single pages take p., e.g., p. B2; multiple pages take pp., e.g., pp. B2, B4 or pp. C1, C3-C4.

Schultz, S. (2005, December 28). Calls made to strengthen state energy policies. *The Country Today*, pp. 1A, 2A.

Single Author

Last name first, followed by author initials.

Berndt, T. J. (2002). Friendship quality and social development. *Current Directions in Psychological Science*, 11, 7-10.

Two Authors

List by their last names and initials. Use the ampersand instead of "and."

Wegener, D. T., & Petty, R. E. (1994). Mood management across affective states: The hedonic contingency hypothesis. *Journal of Personality and Social Psychology*, 66, 1034-1048.

Three and Seven Authors

List by last names and initials; commas separate author names, while the last author name is preceded again by ampersand.

Kernis, M. H., Cornell, D. P., Sun, C. R., Berry, A., Harlow, T., & Bach, J. S. (1993). There's more to self-esteem than whether it is high or low: The importance of stability of self-esteem. *Journal of Personality and Social Psychology*, 65, 1190-1204.

More than Seven Authors

List by last names and initials; commas separate author names. After the sixth author's name, use an ellipses in place of the author names. Then provide the final author name. There should be no more than seven names.

Miller, F. H., Choi, M. J., Angeli, L. L., Harland, A. A., Stamos, J. A., Thomas, S. T., . . . Rubin, L. H. (2009). Web site usability for the blind and low-vision user. *Technical Communication*, 57, 323-335.

Organization as Author

American Psychological Association. (2003).

Unknown Author

Merriam-Webster's collegiate dictionary (10th ed.).(1993). Springfield, MA: Merriam-Webster.

NOTE: When your essay includes parenthetical citations of sources with no author named, use a shortened version of the source's title instead of an author's name. Use quotation marks and italics as appropriate. For example, parenthetical citations of the source above would appear as follows: (Merriam-Webster's, 1993).

Two or More Works by the Same Author

Use the author's name for all entries and list the entries by the year (Earliest comes first).

Berndt, T. J. (1981).

Berndt, T. J. (1999).

When an author appears both as a sole author and, in another citation, as the first author of a group, list the one-author entries first.

Berndt, T. J. (1999). Friends' influence on students' adjustment to school. *Educational Psychologist*, 34, 15-28.

Berndt, T. J., & Keefe, K. (1995). Friends' influence on adolescents' adjustment to school. *Child Development*, 66, 1312-1329.

References that have the same first author and different second and/or third authors are arranged alphabetically by the last name of the second author, or the last name of the third if the first and second authors are the same.

Wegener, D. T., Kerr, N. L., Fleming, M. A., & Petty, R. E. (2000). Flexible corrections of juror judgements: Implications for jury instructions. *Psychology, Public Policy, and Law*, 6, 629-654.

Wegener, D. T., Petty, R. E., & Klein, D. J. (1994). Effects of mood on high elaboration attitude change: The mediating role of likelihood judgements. *European Journal of Social Psychology*, 24, 25-43.

Two or More Works by the Same Author in the Same Year

If you are using more than one reference by the same author (or the same group of authors listed in the same order) published in the same year, organize them in the reference list alphabetically by the title of the article or chapter. Then assign letter suffixes to the year. Refer to these sources in our essay as they appear in your reference list, e.g.: "Berndt (1981a) makes similar claims..."

Berndt, T. J. (1981a). Age changes and changes over time in prosocial intentions and behavior between friends. *Developmental Psychology*, 17, 408-416.

Berndt, T. J. (1981b). Effects of friendship on prosocial intentions and behavior. *Child Development*, 52, 636-643.

References Cited

(All documents must be classified as either: books, periodical articles, published & unpublished researchers or retrieved in internet)

Gomez, Kwanchai A. and Arturo A. Gomez. 1984. *Statistical Procedures for Agricultural Research*. Second Edition. New York: John Wiley and Sons.

Website

McDonald's Corporation. "McDonald's Happy Meal Toy Safety Facts." Retrieved, July 19, 2008. <http://www.mcdonalds.com/corp/about/factsheets.html>.

(Just for Information Purposes)

Chapter V Summary, Conclusions and Recommendations

Literature Cited (bibliography)-are list of materials that helped the researcher in developing the proposal. Not all of these materials may have been mentioned in the body of the proposal. Literature Cited/References Cited- list of materials that were cited or mentioned in the proposal. There are two styles of writing References Cited/Bibliography:

General Format: Authors' Name, Title of Work . Publication Data

Style A: (History & Literature)

Valbuena, Victor T. "Children's Television in Asia: An overview." *Media Asia* 18 (1991): 183-190

Style B: (Social and Natural Sciences)

Valbuena, Victor T. 1991. Children's Television in Asia: An overview. Media Asia 18: 183-190

Valbuena, Victor T. 1991. Children's Television in Asia: An overview. Media Asia 18: 183-190

Gomez, Kwanchai A. and Arturo A. Gomez. 1984. Statistical Procedures for Agricultural Research. Second Edition. New York. John Wiley and Sons.

Ables, Higinio A.; Arabella C. Miranda and Sylvia S. Mateo. 1983. "Asean News Coverage in Three Philippines Daillies." Devcom Quarterly 1:32-40

Website

A citation to website content can often be limited to a mention in the text or in a note (As of July 19, 2008, the McDonald's Corporation listed on its website. . .) if a more formal citation is desired, it may be styled as in the examples below. Because such content is subject to change, include an access date or, if available, a date that the site was last modified. (Chicago Manual of Style, 2010)

1. "Google Privacy Policy," last modified March 11, 2009, <http://www.google.com/intl/en/privacypolicy.html>.
2. "McDonald's Happy Meal Toy Safety Facts," McDonalds Corporation, accessed July 19, 2008, <http://www.mcdonalds.com/corp/about/factsheets.html>.

Google."Google Privacy Policy. "last modified March 11, 2009. <http://www.google.com/intl/en/privacypolicy.html>.
McDonald's Corporation. "McDonald's Happy Meal Toy Safety Facts" Accessed July 19, 2008. <http://www.mcdonalds.com/corp/about/about/factsheets.html>.